Mechanics - Deadline: 3 October 2026, 23:59

Task:

Create a complete, replicated 3D CAD model of the main robot of the Greek team "Hyperion Robotics", which they built for the 2025 edition of Eurobot (videos are available on YouTube).

Important Specifications and Remarks:

1. Project Structure

1. Organize the project files into folders.
2. Add a README file explaining the structure of the project.
3. Use clear, understandable names for parts and folders.

2. Criteria

1. Since the parts will be 3D printed, they must be designed to be "printable", which means:
    1. Minimal use of supports.
    2. Minimal design complexity, to avoid printing problems.
2. If you add anything to the design, make sure it complies with the Eurobot rules and specifications.

3. Electrical Part

Since the electrical part is hidden and you may lack electrical knowledge, follow these specific instructions:

1. **Components:**
    1. Lidar: RPLIDAR A2M12 360°
    2. Motor: Model it as a cylinder with a radius of 20 mm and a length of 120 mm, with a 16 mm shaft.
    3. Wheels: Model them as cylinders with a radius of 35 mm and a length of 30 mm.
    4. Raspberry Pi 5 (Optional)
    5. Two electronic boards (approximate each as a 100 mm × 100 mm × 20 mm rectangular cuboid) (Optional)
    6. Battery: LiPo 3S, approximately 155 mm × 45 mm × 34 mm
    7. Stepper motor: NEMA 17 42HS34-1334AC
    8. Servo motor: you may use either the MG996R or the Micro Servo 9g.
    9. Electromagnets: simply approximate them.
    10. Emergency stop button: red, Ø 40 mm head, Ø 22 mm mounting hole.

2. **Structure:**
    1. All electronics, except the battery, must be housed in a drawer.
    2. The drawer must be able to be fully deployed, so that the electrical team can easily replace their components.
    3. The battery must have its own box to protect it from external impacts.
    4. Take cable management into consideration, i.e. how and where the cables are routed. (Don't worry, we will take your lack of electrical knowledge into account.)

4. Other

1. Do not model the game elements or the rectangular box with the ArUco code on top of the robot (an ArUco code is similar to a QR code).
2. For the tin can gripper, Hyperion Robotics used an electromagnet.

N.B.:

1. Upload your work to a GitHub repository.
2. If you have any questions, don't be afraid to ask in the group chat.
